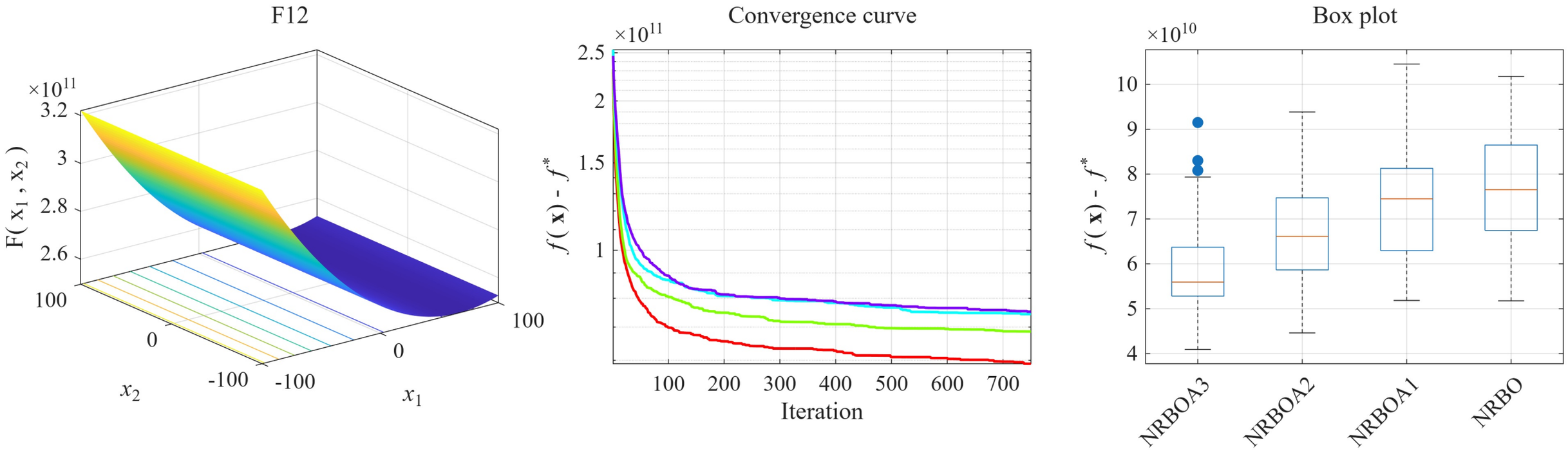

F12
×10^11
3.2
3
2.8
2.6
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
×10^11
2.5
2
1.5
1
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^10
10
9
8
7
6
5
4
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO